Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 1 – Coordinate Geometry

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find the gradient of the line joining $A(1, 2)$ and $B(5, 10)$. **[2]**

2. Find the gradient of the line joining $A(-1, 3)$ and $B(4, -2)$. **[2]**

3. Find the equation of the line through $(2, 3)$ with gradient 4. **[3]**

4. Find the equation of the line through $(1, -2)$ with gradient $-\frac{1}{2}$. **[3]**

5. Find the equation of the line through $A(1, 4)$ and $B(3, 8)$. **[3]**

6. Find the equation of the line through $A(-2, 1)$ and $B(4, 5)$. **[3]**

7. Find the midpoint of $A(2, 5)$ and $B(6, 1)$. **[2]**

8. Find the midpoint of $A(-3, 4)$ and $B(5, -2)$. **[2]**

9. Find the distance between $A(1, 2)$ and $B(4, 6)$. **[3]**

10. Find the distance between $A(-1, 3)$ and $B(5, -5)$. **[3]**

11. Find the equation of the line parallel to $y = 3x - 1$ passing through $(2, 5)$. **[3]**

12. Find the equation of the line perpendicular to $y = 2x + 3$ passing through $(1, 4)$. **[4]**

13. Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$ through $(3, 1)$. **[4]**

14. Does the point $(2, 7)$ lie on the line $y = 3x + 1$? **[2]**

15. Find the intercepts of the line $2x + 3y = 12$. **[3]**

Section B – Hard (Q16–25)

16. Find the equation of the perpendicular bisector of $A(1, 2)$ and $B(5, 6)$. **[5]**

17. Find the equation of the perpendicular bisector of $A(-2, 3)$ and $B(4, 1)$. **[5]**

18. The line l_1 has equation $y = 2x + 1$. The line l_2 is perpendicular to l_1 and passes through $(4, 3)$. Find the equation of l_2 and the intersection point of l_1 and l_2. **[6]**

19. Find the area of the triangle formed by the lines $y = x$, $y = 0$ and $x + y = 4$. **[4]**

20. Points $A(1, 1)$, $B(4, 5)$, $C(7, 1)$. Show that ABC is isosceles and find its area. **[6]**

21. Find the equation of the line that passes through the intersection of $y = 2x - 1$ and $y = x + 3$ and is parallel to $y = 3x$. **[5]**

22. The points $A(2, 3)$, $B(6, 7)$, $C(8, 3)$. Find the equation of the median from C. **[5]**

23. Find the angle that the line $y = \sqrt{3}x + 1$ makes with the positive x-axis. **[3]**

24. Show that the points $A(1, 2)$, $B(3, 6)$, $C(7, 5)$ form a right-angled triangle. **[5]**

25. Find the coordinates of the point dividing the line segment from $A(1, 2)$ to $B(7, 8)$ in the ratio $1 : 2$. **[4]**

End of Worksheet • *Show all working clearly.*